```
100 'THE INVERSE DISCRETE FOURIER TRANSFORM
110 'The time domain signal, held in XX[ ], is calculated from the frequency domain signals,
120 'held in REX[ ] and IMX[ ].
130 '
140 DIM XX[511]                 'XX[ ]  holds the time domain signal
150 DIM REX[256]                'REX[ ] holds the real part of the frequency domain
160 DIM IMX[256]                'IMX[ ] holds the imaginary part of the frequency domain
170 '
180 PI = 3.14159265             'Set the constant, PI
190 N% = 512                    'N% is the number of points in XX[ ]
200 '
210 GOSUB XXXX                  'Mythical subroutine to load data into REX[ ] and IMX[ ]
220 '
230
240 '                           'Find the cosine and sine wave amplitudes using Eq. 8-3
250 FOR K% = 0 TO 256
260   REX[K%] =  REX[K%] / (N%/2)
270   IMX[K%]  = -IMX[K%] / (N%/2)
280 NEXT K%
290 '
300 REX[0] = REX[0] / 2
310 REX[256] = REX[256] / 2
320 '
330 '
340 FOR I% = 0 TO 511           'Zero XX[ ] so it can be used as an accumulator
350   XX[I%] = 0
360 NEXT I%
370 '
380 '                           Eq. 8-2  SYNTHESIS METHOD #1.  Loop through each
390 '                           frequency generating the entire length of the sine and cosine
400 '                           waves, and add them to the accumulator signal, XX[ ]
410 '
420 FOR K% = 0 TO 256           'K% loops through each sample in REX[ ] and IMX[ ]
430   FOR I% = 0 TO 511    'I% loops through each sample in XX[ ]
440     '
450     XX[I%] = XX[I%] + REX[K%] * COS(2*PI*K%*I%/N%)
460     XX[I%] = XX[I%] + IMX[K%] * SIN(2*PI*K%*I%/N%)
470     '
480   NEXT I%
490 NEXT K%
500 '
510 END
```

Alternate code for lines 380 to 510

```
380 '                           Eq. 8-2  SYNTHESIS METHOD #2.  Loop through each
390 '                           sample in the time domain, and sum the corresponding
400 '                           samples from each cosine and sine wave
410 '
420 FOR I% = 0 TO 511           'I% loops through each sample in XX[ ]
430   FOR K% = 0 TO 256         'K% loops through each sample in REX[ ] and IMX[ ]
440     '
450     XX[I%] = XX[I%] + REX[K%] * COS(2*PI*K%*I%/N%)
460     XX[I%] = XX[I%] + IMX[K%] * SIN(2*PI*K%*I%/N%)
470     '
480   NEXT K%
490 NEXT I%
500 '
510 END
```

TABLE 8-1